

SOLUTIONS

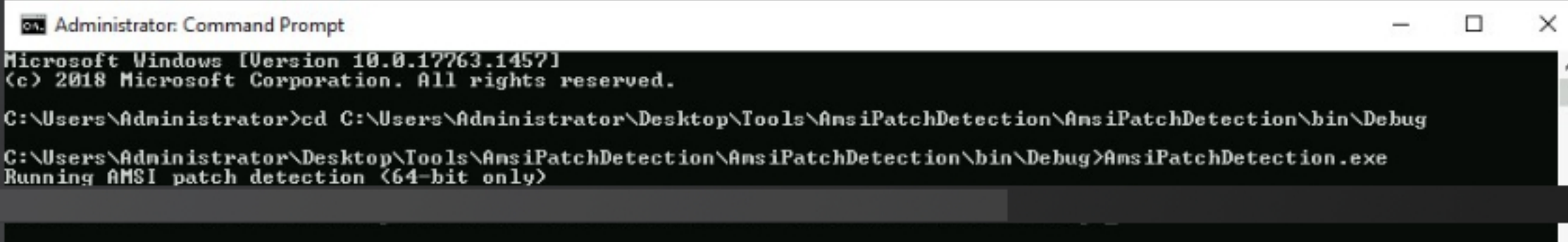
Below, you can find solutions for each task. Remember though, that you can follow your own strategy, which may be different from the one explained in the following lab.

Task 1. Hunt for AMSI bypasses

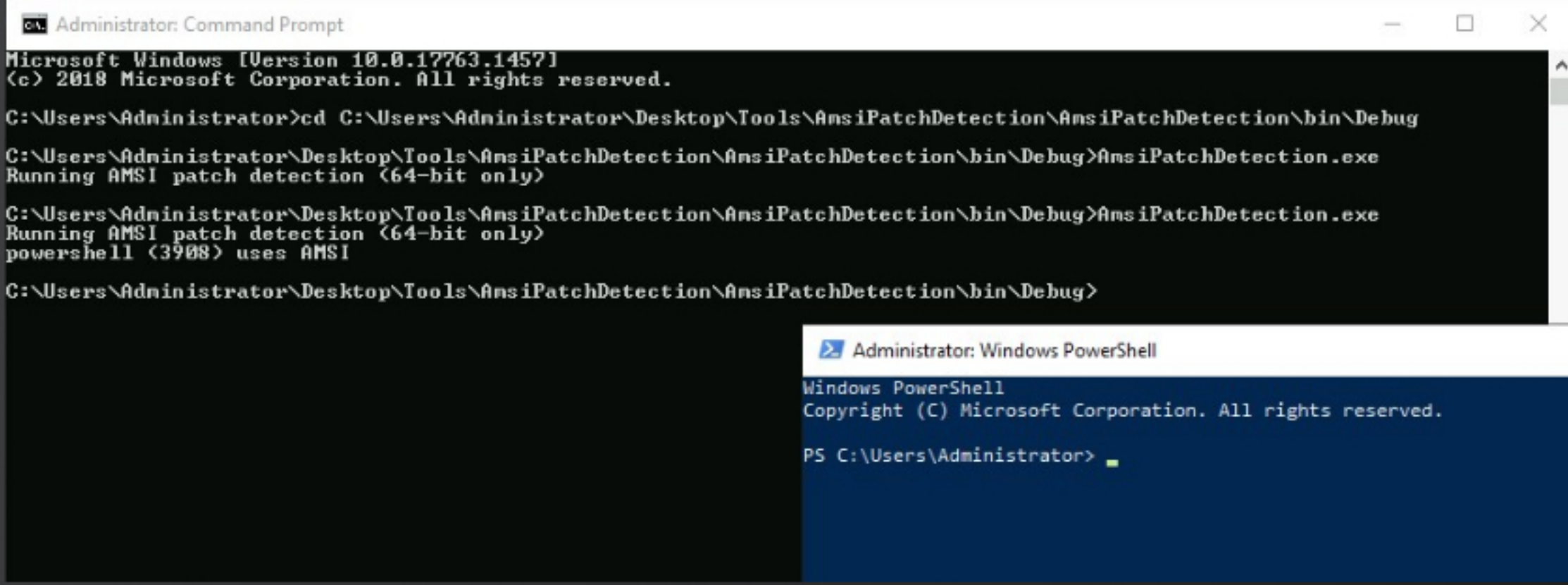
To hunt for the AMSI bypass using the provided code (*ASBBypass.ps1*), execute the following.

- Execute *cmd.exe* as an administrator
- Execute the AMSI bypass detection executable located at *C:\Users\Administrator\Desktop\Tools\AmsiPatchDetection\AmsiPatchDetection\bin\Debug*

You should see something similar to the below.



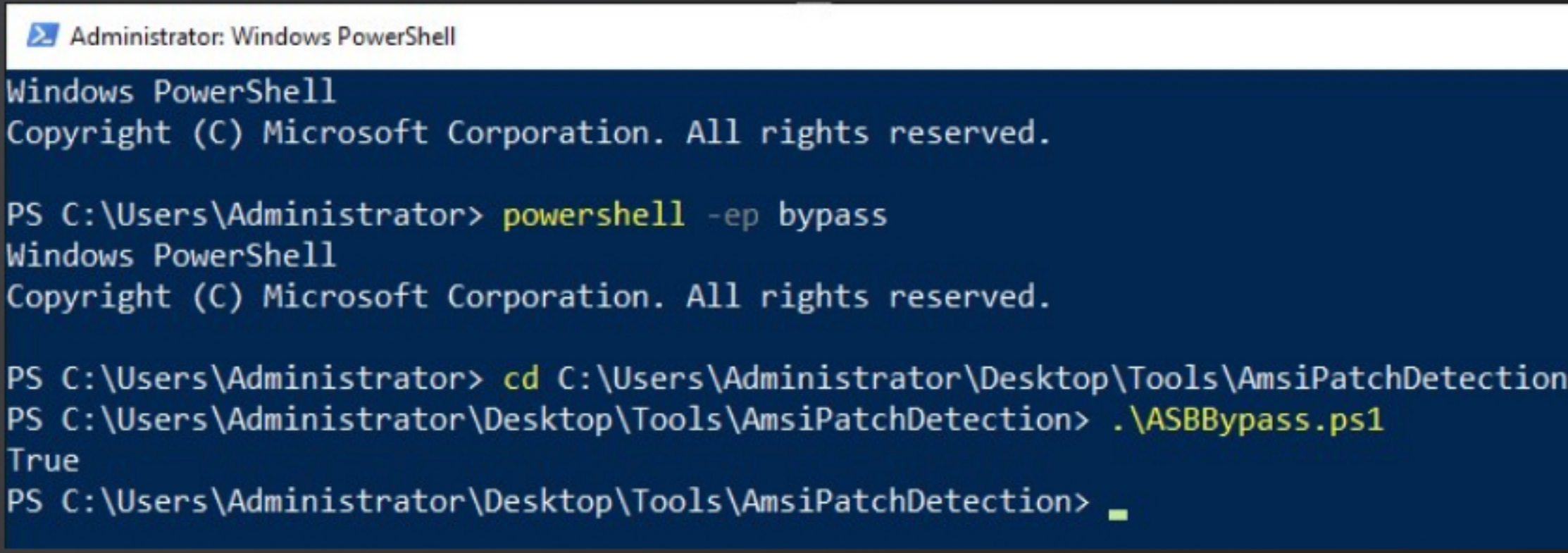
- Execute powershell.exe and execute the detection executable once again. You should see something similar to the below (PowerShell uses AMSI).



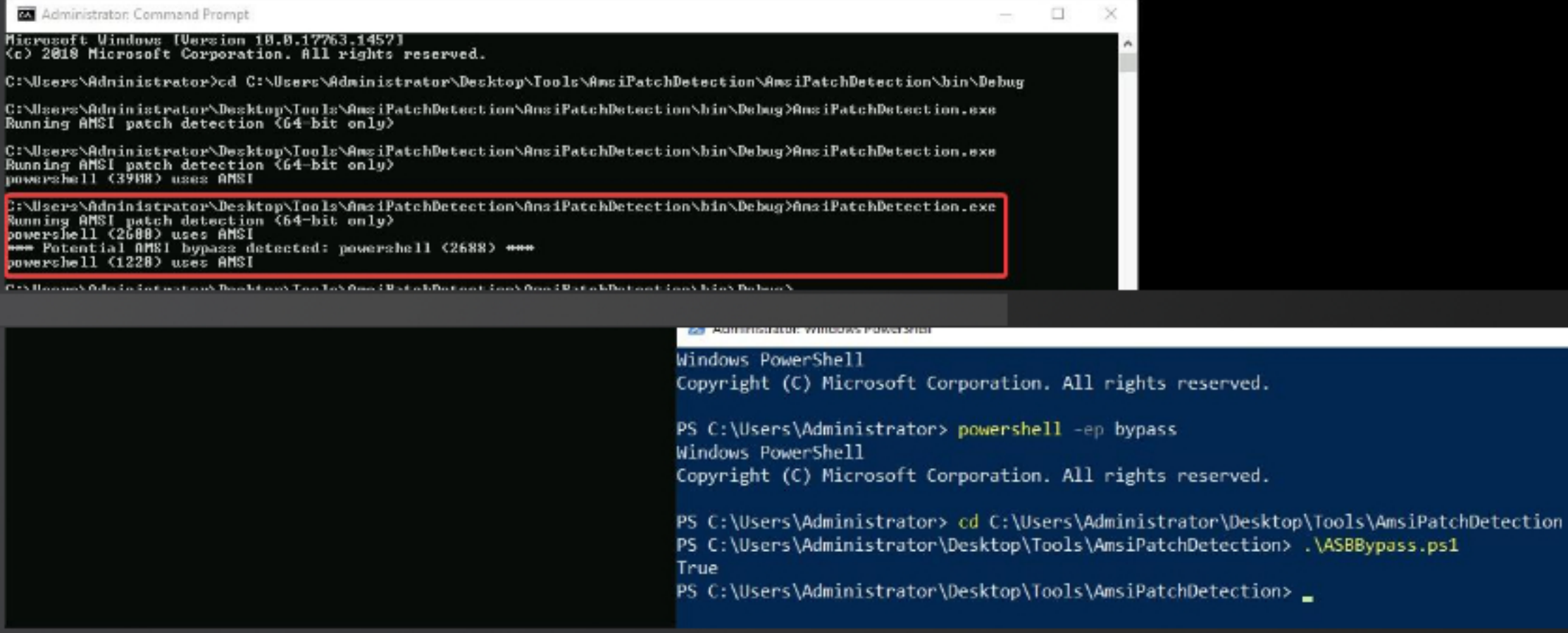
- Emulate the AMSI bypass code (*ASBBypass.ps1*) located in the *Downloads* directory, by executing the below inside the PowerShell terminal you previously spawned.

```
powershell -ep bypass

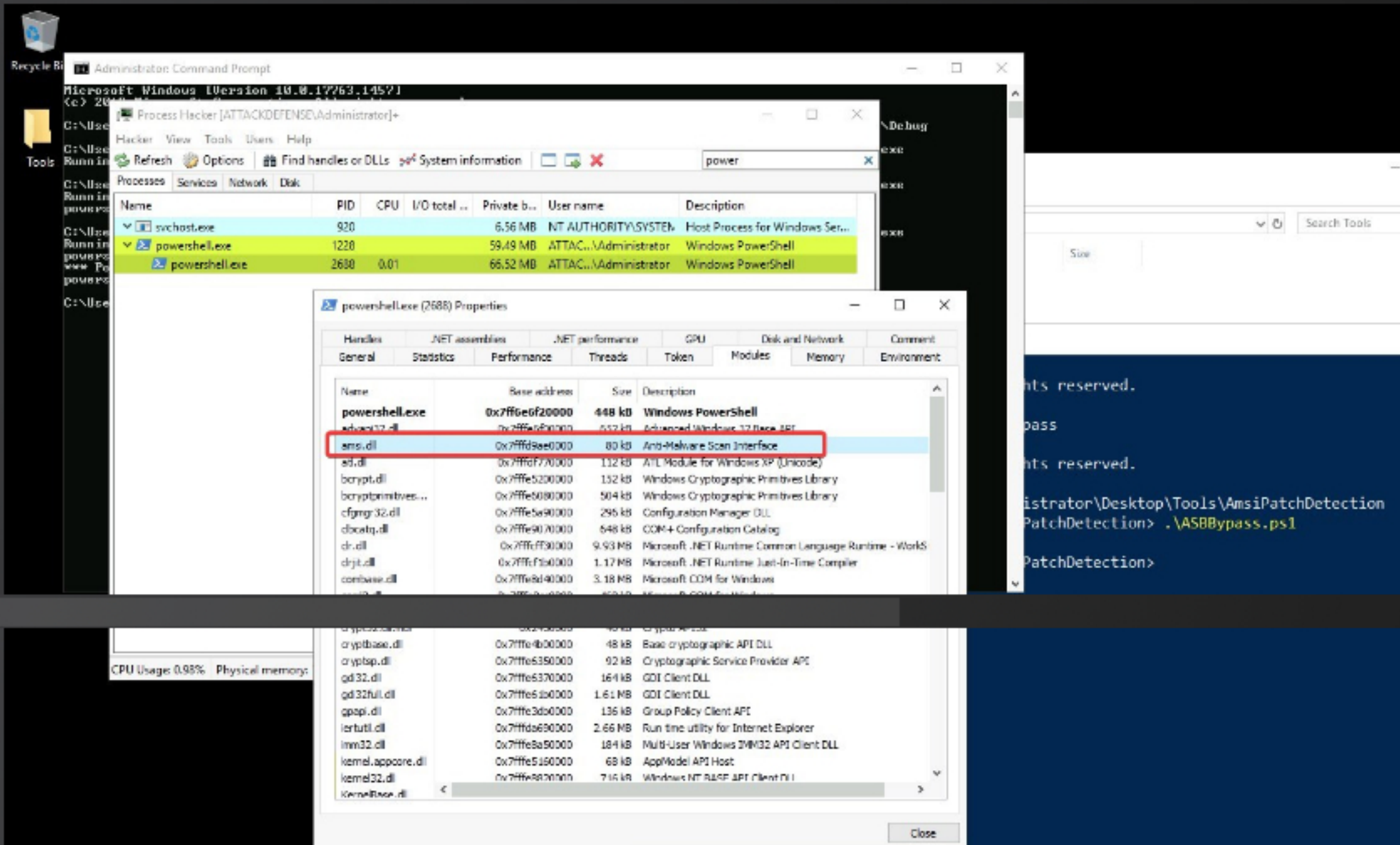
.\ASBBypass.ps1
```



- Finally, execute the AMSI bypass detection executable again. You should now see it detecting the AMSI bypass on the PowerShell process.



If you start *Process Hacker 2* as an administrator, you will also be able to see the memory location of *amsi.dll* having read, write and execute rights (this is not a default behavior, it was caused by the AMSI bypass code).

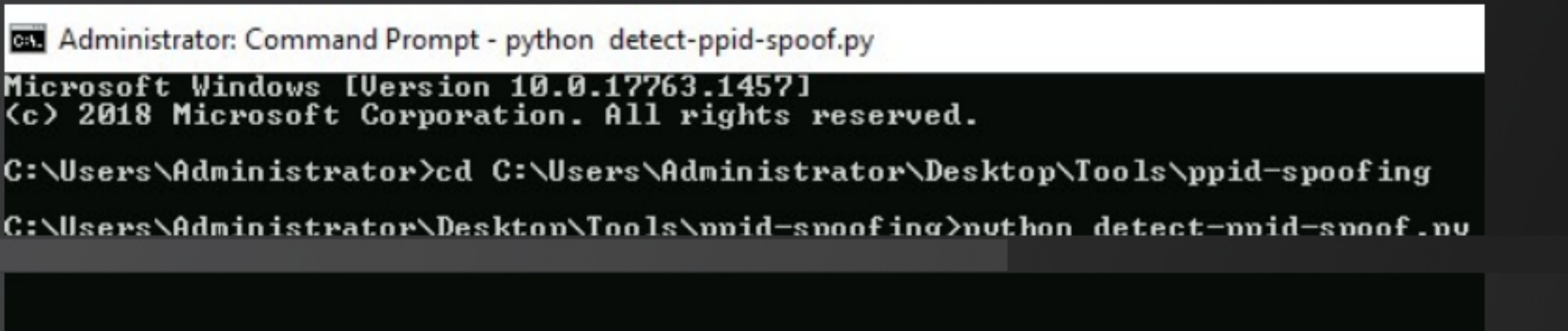


Task 2. Hunt for Parent PID Spoofing

To hunt for PPID Spoofing using the provided code (*PPID-Spoof.ps1*), execute the following.

- Execute *cmd.exe* as an administrator
- Execute the PPID Spoofing detection script (*detect-ppid-spoof.py*) located at *C:\Users\Administrator\Desktop\Tools\ppid-spoofing*

You should see something similar to the below.



- Emulate the attack as follows.
 - Use the task manager and note *explorer.exe*'s PID
 - Execute *powershell.exe* as an administrator and then, execute the below

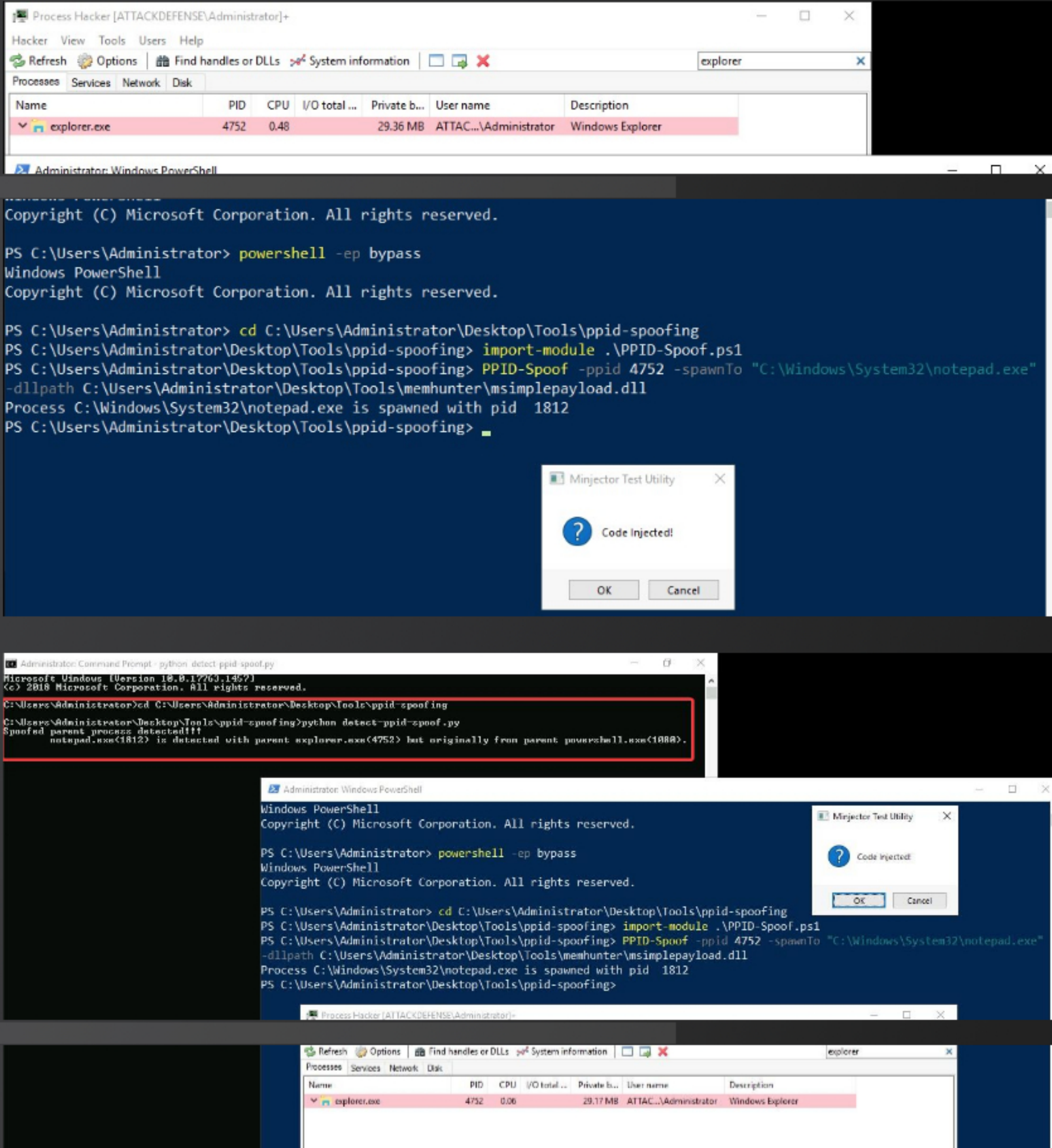
```
powershell -ep bypass

cd C:\Users\Administrator\Desktop\Tools\ppid-spoofing

import-module .\PPID-Spoof.ps1

PPID-Spoof -ppid 4752 -spawnTo "C:\Windows\System32\notepad.exe" -dllpath C:\Users\Ad
```

You should see the below.



The detection script successfully identified the PPID spoofing activity.

References

1. <https://ionize.com.au/detecting-amsi-bypass/>
2. <https://blog.f-secure.com/detecting-parent-pid-spoofing/>